Brain & Language 269 (2025) 105617

Contents lists available at ScienceDirect

Brain and Language

journal homepage: www.elsevier.com/locate/b&l

Research Article

Modulating language and executive functions in bilingual aphasia with cerebellar tDCS: a case series

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ARTICLE INFO

Keywords:
Primary Progressive Aphasia
Aphasia
Bilingualism
Cerebellum
transcranial Direct Current Stimulation
Executive Functions

ABSTRACT

This case series explores the effects of cerebellar transcranial direct current stimulation (tDCS) on language and executive functions in bilinguals with aphasia. We present seven Dutch-French bilingual participants diagnosed with Primary Progressive Aphasia (PPA) or post-stroke aphasia, including one non-fluent variant PPA (nfvPPA), three logopenic variant PPA (lvPPA), one semantic variant PPA (svPPA), and two post-stroke non-fluent aphasia patients. 20 min of 2 mA anodal tDCS to the right posterolateral cerebellum was combined with speech and language therapy. We administered subtests of the Bilingual Aphasia Test, Boston Naming Test, Stroop, and Attention Network Tests. Cerebellar tDCS compared to sham led to greater enhancement of language recovery in both languages, and inhibitory control, providing evidence for the cerebellum's role in both language and executive processes. Our findings contribute to a deeper understanding of cerebellar stimulation as a therapeutic tool in bilingual aphasia, with implications for future research and clinical interventions.

1. Introduction

1.1. Language, bilingualism and executive functions

Language processing is thought to rely not only on domain-specific mechanisms but also on domain-general executive functions (EFs) (e.g., Thompson et al., 2018). EFs contribute to various aspects of language use, including sentence comprehension and word retrieval (Mohapatra, 2019; Murray, 2012; Esslinger, 1996). These higher-order cognitive abilities are involved in planning, reasoning, problem solving, and cognitive control (Alvarez & Emory, 2006). EFs are commonly divided into three components: *updating* (i.e., maintaining and manipulating information in working memory), *shifting* (adapting to changing linguistic or task demands), and *inhibitory control* (suppressing irrelevant or interfering information) (Miyake et al., 2000). These components are engaged during language processing (e.g., Daneman & Carpenter, 1980; Martin & Romani, 1994; Green & Abutalebi, 2013; Badre & Wagner, 2007; Kaushanskaya et al., 2017), and their neural underpinnings partly overlap with those of language, involving frontal and prefrontal cortices and their subcortical connections (Bettcher et al., 2016; Stuss, 2011). In bilingual individuals, this relationship becomes more complex. Bilinguals regularly engage in bilingual language control *(*BLC) to manage cross-language interference, particularly through inhibitory mechanisms (Abutalebi and Green, 2007; Green, 1998; Green & Abutalebi, 2013; Declerck et al., 2021). There is evidence that bilinguals rely on control processes to switch between languages, but it remains debated to what extent these control mechanisms overlap with domain-general executive functions: a question central to the so-called "bilingual advantage debate" (e.g., Green & Abutalebi, 2013; Grundy, 2020; Lehtonen et al., 2018; van den Noort et al., 2019). Neuroimaging studies

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https://doi.org/10.1016/j.bandl.2025.105617
Received 3 February 2025; Received in revised form 11 June 2025; Accepted 3 July 2025
Available online 16 July 2025

suggest that bilingualism may lead to structural and functional changes not only in traditional language areas, but also in executive functions regions (Mechelli et al., 2004; Pliatsikas et al., 2015; Ressel et al., 2012; Wartenburger et al., 2003). The extent and nature of these effects vary with factors such as age of second language (L2) acquisition (AoA), L2 exposure and proficiency (Bialystok et al., 2012; Luk & Bialystok, 2013; Perani et al., 1998; Pliatsikas et al., 2017). Despite growing research, the relationship between bilingualism and executive functioning remains incompletely understood.

1.2. Aphasia: a disruption of language

Aphasia refers to acquired impairments in language processing due to brain injury or degeneration, affecting a person's ability to understand, produce, or repeat language across modalities (speaking, listening, reading, and writing). primary progressive aphasia (PPA) is a neurodegenerative syndrome primarily impacting language abilities, often preceding broader cognitive decline. PPA is classified into three main variants: non-fluent/agrammatic (nfvPPA), logopenic (lvPPA) and semantic (svPPA) PPA, based on distinct clinical, linguistic and imaging features (Gorno-Tempini et al., 2011). The nfvPPA variant is marked by effortful, halting speech, agrammatism, and, frequently, apraxia of speech. Comprehension of syntactically complex sentences can be impaired, while single-word comprehension and object knowledge tend to be preserved. Atrophy typically occurs in the left posterior fronto-insular region and is most often associated with frontotemporal lobar degeneration (FTLD) with tau pathology (Gorno-Tempini et al., 2011; Taylor et al., 2024). The lvPPA variant is characterized by word-finding difficulties and impaired repetition of sentences and phrases, along with phonological errors in speech. Comprehension of single words and object knowledge are usually spared. Imaging typically shows atrophy in the left posterior perisylvian or parietal region, and this variant is most commonly associated with Alzheimer's disease pathology (Gorno-Tempini et al., 2011; Taylor et al., 2024).

In svPPA, the core symptoms include impaired confrontation naming and single-word comprehension, often accompanied by surface dyslexia or dysgraphia and degraded object knowledge. Speech production and repetition are generally preserved. This variant is linked to prominent atrophy in the anterior temporal lobes and is most often associated with FTLD-TDP pathology (Gorno-Tempini et al., 2011; Taylor et al., 2024).

Pharmacological interventions are limited, and speech and language therapy (SLT) currently remains the primary tool for preserving language abilities (Wauters et al., 2023).

Post-stroke aphasia affects up to one-third of stroke survivors (Engelter et al., 2006; Flowers et al., 2017), and presents with fluent (receptive) or non-fluent (expressive) profiles depending on the lesion site (Sheppard & Sebastian, 2021). While both post-stroke aphasia and PPA involve language impairments, they differ in onset, pathology and neural mechanisms of recovery, requiring tailored therapeutic approaches.

1.3. Bilingual aphasia

Bilingual individuals vary in language proficiency, dominance, and use, which adds complexity to how their two language systems are organized and managed. This has raised questions about the cognitive and neural underpinnings of bilingualism, including how shared and distinct networks support each language and how control mechanisms manage cross-language interference (Del Maschio & Abutalebi, 2019). In bilingual aphasia, these complexities are amplified by damage affecting one or both languages, often resulting to varied patterns of impairment, decline and recovery. Recovery may occur in parallel across languages or differentially, with one language improving more than the other (Paradis, 1977; 2001, Fabbro, 2001; Costa et al., 2019). Some studies suggest that impairments may stem from disrupted language control—such as difficulty inhibiting the non-target language—rather than from damage to language-specific areas, as seen in pathological switching or mixing (Abutalebi and Green, 2007; Fabbro, 2001, Green et al., 2004; Stemmer & Whitaker, 1998; Van der Linden et al., 2018). These issues also affect clinical decision-making. In bilingual aphasia, managing multiple language systems and facilitating cross-linguistic transfer of treatment effects are central therapeutic goals (DeLuca et al., 2020, Yuan et al., 2022). Speech and language therapists (SLTs) must take into account factors such as language dominance and inhibitory control, which shape treatment outcomes and the likelihood of transfer between languages (Edmonds & Kiran, 2006; Kiran & Roberts, 2010; Kiran et al., 2013). Although no clear consensus exists on whether to prioritize the dominant or non-dominant language, evidence suggests that both can be effective, with treatment choices best guided by individual profiles (Goral et al., 2023).

1.4. Transcranial direct current stimulation (tDCS)

tDCS is a non-invasive neuromodulatory technique that delivers a low-intensity electrical current (usually 1–2 mA) through scalp electrodes to modulate cortical excitability (Priori et al., 1998). Depending on the polarity, anodal stimulation is generally associated with increased excitability, while cathodal stimulation tends to reduce excitability. These shifts in excitability are thought to influence synaptic plasticity and cortical reorganization. Importantly, tDCS effects are not limited to the stimulated region but can extend to functionally and structurally connected networks, as evidenced by neuroimaging studies (e.g., Sebastian et al., 2017, Ficek et al., 2018; Almeida et al., 2023). tDCS has been used as a modulatory tool in healthy individuals, but results are inconsistent and often limited in scope, particularly for language functions (e.g., Westwood et al., 2017; Klaus & Hartwigsen, 2020). Potential benefits of tDCS may be more robust in clinical populations with existing brain damage, where neuromodulation may support recovery-related plasticity. Beyond aphasia, tDCS is also being investigated as a treatment approach for other neurological and psychiatric conditions (e.g., pain treatment (Pacheco-Barrios et al., 2020), depression (Brunoni et al., 2016), autism spectrum disorder (García-González et al., 2021), where modulation of dysfunctional networks may offer therapeutic benefit.

In aphasia, tDCS is currently being explored as a research tool to enhance treatment outcomes. It is most often used in combination with SLT, in a paired approach designed to promote greater or more lasting treatment gains. tDCS has shown promise in improving language abilities in both post-stroke aphasia (see Marangolo, 2020 for review) and PPA (see Coemans et al., 2021 for review). In post-stroke aphasia, common stimulation targets include left hemisphere language regions such as the inferior frontal gyrus (IFG) and superior temporal gyrus (STG), as well as the right hemisphere to modulate compensatory or maladaptive activity (Hamilton et al., 2011; Biou et al., 2019). In PPA, studies frequently focus on the left hemisphere language areas, most often the IFG (see Coemans et al., 2021 for a review), though the heterogeneity of the disorder and variability in outcomes highlight the need for further research. Given these uncertainties, the search for alternative or complementary stimulation sites is ongoing.

1.5. The (linguistic) cerebellum and cerebellar tDCS

The cerebellum, traditionally associated with motor control, is increasingly recognized for its contribution to higher-order functions, including language and executive processing (e.g., Van Overwalle, 2024). Neuroimaging and lesion studies have implicated the cerebellum in various language tasks such as verbal fluency, verb generation, sentence construction, sentence comprehension, and phoneme discrimination (e.g., Geva et al., 2021; Leggio et al., 2020; Mariën et al., 2017). Furthermore, cerebellar activation (Szaflarski et al., 2013) and the integrity of corticocerebellar white matter tracts (Keser et al., 2023) have been linked to better rehabilitation outcomes in aphasia. While cerebellar tDCS has shown promise in enhancing language outcomes in post-stroke aphasia (Marangolo et al., 2018; Sebastian et al., 2020; Sebastian et al., 2017), to date no published studies have investigated cerebellar tDCS in PPA or bilingual aphasia (except from our own case reports: Coemans et al., 2023; Coemans et al., 2024). The cerebellum may represent a promosing neuromodulatory target, given its dense connectivity with cortical language areas (Jobson et al., 2022; Schmahmann, 2019; Stoodley, 2012) and its involvement in executive functions and language control (Abutalebi & Green, 2016, Yuan et al., 2022). Further, the cerebellum is often relatively spared in both neurodegenerative and vascular aphasia (Lombardi et al., 2021). This facilitates more accurate modeling of current flow and allows for clearer attribution of observed effects to cerebellar function and its supratentorial connections.

Considering the sparsity of intervention studies in bilingual aphasia, especially in the context of progressive conditions like PPA, but also in stroke-related aphasia (Goral et al., 2023), and the potential opportunities of cerebellar tDCS as a therapeutic adjunct, this study aims to evaluate the efficacy of cerebellar tDCS in bilingual individuals with aphasia. Specifically: (1) We ask whether cerebellar tDCS combined with language therapy improves language outcomes in the treated language, the non-treated language, or both. (2) We explore whether the effects differ across aphasia subtypes. (3) We aim to identify which specific linguistic functions benefit from cerebellar tDCS, and lastly (4) we evaluate whether executive functions are influenced, particularly those related to language control (i.e., inhibition).

2. Methods

This section gives an overview of the methods and materials. SLT details are provided in each participant's individual section.

2.1. Study design

This study employed a within-subject, double-blinded, randomized design (Sebastian et al., 2020) (Fig. 1). Participants received both anodal cerebellar tDCS and sham tDCS in randomized order. Each stimulation phase was paired with functionally oriented SLT targeting individual language deficits, including a personalized oral naming task. Sham tDCS mimicked real stimulation through brief current ramping. Language and executive functions were assessed pre- and post-treatment, with a 2-month follow-up.

2.2. Participants

Participants (n = 7), were Dutch-French bilinguals with a confirmed diagnosis of PPA or post-stroke aphasia, recruited in Belgium between 2021 and 2023. Inclusion criteria included fluency in both languages, right-handedness, and absence of neurological disorders other than aphasia. Exclusion criteria included severe cognitive impairment, psychiatric disorders, or contraindications for non-invasive brain stimulation. Years of education were calculated as the sum of 6 years of secondary education and any further education. The final sample consisted of five participants with PPA and two with post-stroke aphasia.

2.3. Data collection

Language and executive functions were assessed before and after each tDCS phase. Testing sessions lasted a total of 4–5 hours and were split across days to avoid fatigue. Each language was assessed on a separate day.

2.4. Materials

2.4.1. Bilingual language use, diagnostic and untrained language tasks

The LEAP-Q was used to document bilingual language use, including language abilities, age of acquisition (AoA), usage patterns and self-rated proficiency (Marian et al., 2007). The **Bilingual Aphasia Test (BAT)** was administered in both languages, with subtests chosen based on the patient's difficulties, to limit assessment time (Paradis & Libben, 2014).

2.4.2. Trained language task

As therapy included picture naming supported by Semantic Feature Analysis (SFA; Boyle & Coelho, 1995), a **personalized naming task** was developed. This task consisted of 100 picture items matched to the Boston Naming Test (BNT) in word length and frequency. Pictures misnamed in at least one of the two baseline assessments were selected for training. Scoring followed the BNT format.

The **BNT** was administered in Dutch and French to assess generalization to the untreated language. It uses a 4-point scale (0,1,2,3), based on accuracy and reaction time (Roomer et al., 2011), with a maximum total score of 177 (59 picture items multiplied by 3 points; (Kaplan et al., 1983).

2.4.3. General cognitive (non-linguistic) assessment

The **Mini-Mental State Examination (MMSE)** was used to measure global cognitive function. Scores range from 0 to 30, with lower scores indicating greater cognitive impairment (Folstein et al., 1975).

The **Attention Network Test (ANT) and Stroop Test** were administered to assess inhibitory control. The non-verbal ANT evaluates executive network efficiency by measuring the Flanker effect, defined as

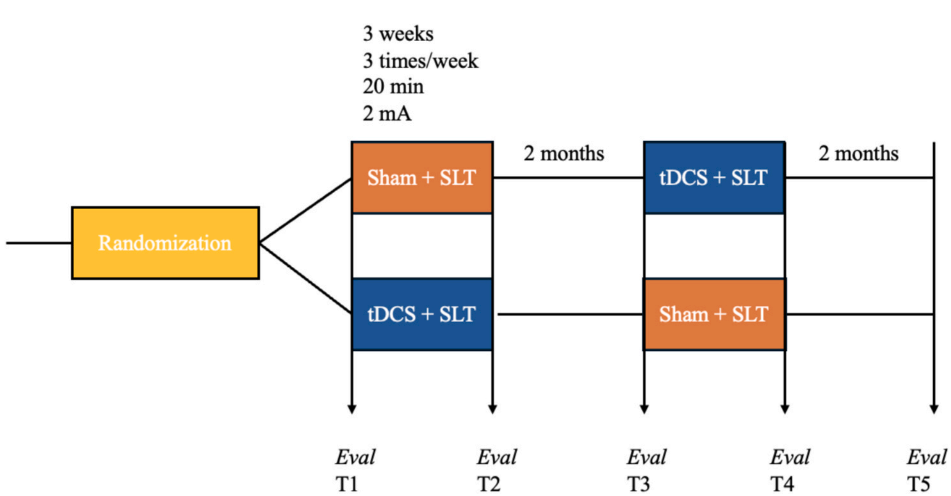

Fig. 1. Study Design.

the difference in reaction times between congruent and incongruent trials. A larger Flanker effect indicates greater difficulty inhibiting interference (Coemans et al., 2023). The Stroop Test (Stroop, 1935) was administered in the participant's dominant language, using adapted Dutch (Hammes, 1971) and French (Godefroy & Editions Solal, 2007) versions. The interference score is calculated by comparing reaction times on incongruent (Card 3) and congruent (Card 2) conditions. Higher scores reflect greater interference. Age- and education-based adjustments were applied using language-specific norms.

2.5. Intervention

tDCS (Oasis Pro, Mind Alive Inc., Canada) was administered at 2 mA using 3x5 cm sponge electrodes, with the anode placed over the right posterolateral cerebellum (lobule VII/Crus II), 1 cm below and 4 cm lateral to the inion (Sebastian et al., 2020), and the cathode on the right deltoid (Sebastian et al., 2020). Stimulation was delivered concurrently with SLT, with each session comprising 20 min of tDCS and 20–40 min of personalized SLT, three times a week across 3 weeks per phase, separated by a 2-month interval (Fig. 1). SLT included functionally oriented therapy, based on standard clinical care, and personalized naming therapy targeting items misnamed during baseline assessments (see Section 2.4). Naming therapy included SFA (Boyle & Coelho, 1995), semantic and phonological cueing, and repetition of clinician-named items.

Sham stimulation was delivered using the Oasis Pro's built-in mode, which mimics the sensation of active tDCS (e.g., tingling) for 30–60 s before ramping down to a sub-threshold level (<0.1 mA) that produces no neuromodulatory effects. This method supports participant blinding. To ensure experimenter blinding, the device was set to mode "1″ or "2" for each session, with the actual condition (active or sham) pre-coded by the manufacturer and revealed only after data collection. Thus, both participants and experimenters remained blind to the stimulation condition.

2.6. Data analysis

Statistical analysis compared pre- and post-treatment scores for each stimulation type (sham and tDCS), as well as pre-treatment and follow-up scores per individual. Naming scores were analyzed using McNemar's test (McNemar, 1947), and the Flanker effect was assessed using repeated measures ANOVA with within-subject factors: "congruency type" (congruent vs. incongruent), "stimulation type" (sham vs. anodal) and "timing" (pre-treatment, post-treatment and follow-up), followed by Bonferroni-corrected main effects and pairwise comparisons.

3. Results

The results section presents data per participant, including clinical profile, language background, and the effects of SLT with sham and anodal cerebellar tDCS. Table 1 details all participants' language background and trained SLT tasks. Table 2 is a summary table that indicates whether language and inhibitory control performance improved (green), declined (red), or remained stable (yellow) following sham and anodal tDCS. Detailed results are presented per case. See Supplementary tables 2-8 for all individual pre-and posttreatment test scores.

1. **Cases WITH PRIMARY PROGRESSIVE APHASIA**

3.1. CASE 1 – nfvPPA

3.1.1. Patient Characteristics and language background

Mr. R., a 69-year-old male diagnosed with nfvPPA and Apraxia of Speech (AoS), 2 years prior to study enrollment, presented with speech and language features consistent with diagnostic criteria (Gorno-Tempini et al., 2011; Utianski et al., 2018). These included impaired articulation, disrupted prosody, and inconsistent articulatory errors (e.g., distortions, deletions, substitutions), particularly in longer words and consonant clusters. His speech was slow and effortful, and he made syntactic errors, although there were no reported memory or attention issues. Since diagnosis, **SLT had been delivered in his mother tongue (L1)**, Dutch, focusing on exercises such as picture description, anagrams, sentence construction/completion. Upon inclusion, therapy was supplemented by a personalized picture naming task of 40 items. Mr. R. had learned French (L2) from age six, pursued his education in French, and later worked as a Dutch-French language teacher. He considered his proficiency in French equal to Dutch (10/10) and maintained his language skills through friendships and pen-pal exchanges. He primarily spoke Dutch at home.

3.1.2. Baseline performance

Given Mr. R.'s pronounced AoS and slow, non-fluent speech, the BAT screening test version was used to reduce assessment time. For the ANT, no normative data specific to PPA populations were available. Instead, we reference Fuentes et al. (2010), who reported mean reaction times for 18 monolingual Alzheimer's patients (average age: 72): 992 ms in the congruent condition and 1153 ms in the incongruent condition.

Table 1
Summary Language Background and SLT.

	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
Aphasia type	nfvPPA	lvPPA	lvPPA	lvPPA	svPPA	NFPS	NFPS
Sex, age	M, 69	M, 59	M, 63	M, 73	M, 77	M, 73	F, 69
L1*	Dutch	French	Dutch	Both	French	French	French
AoA	6	12	2	Birth	Birth	Birth	6
Pre-diagnosis language proficiencies (self-report)	10/10 (L1 and L2)	10/10 (L1 and L2)	10/10 (L1 and L2)	10/10 (L1 and L2)	10/10 (L1 and L2)	L1 10/10; L2 9/10	L1 8/10; L2 5/10
Patterns language impairment	Differential, L1 better than L2	Parallel decline	Differential, L2 better than L1	Parallel decline	Differential, L1 better than L2	Differential, L1 better than L2	Differential, L1 better than L2
SLT language	L1	L2 (dominant)	L2 (dominant)	French	L1	L2 (non-dominant)	L1
SLT tasks	PN, picture description, sentence completion, anagrams, sentence construction	PN, semantic categories, word-finding exercises, syllable reinforcement	PN, phonological component analysis, metaphonology training, syllable reinforcement	PN, semantic categories, verbal fluency, memory games	PN, semantic categories, verbal fluency	PN, word-finding exercises, e.g., verbal fluency	PN, sentence comprehension

Note. Abbreviations: nfvPPA = non-fluent variant PPA, AoS = Apraxia of Speech, lvPPA = logopenic variant PPA, svPPA = semantic variant PPA, NFPS = non-fluent post-stroke, AoA = Age of Acquisition of L2, SLT= Speech and Language Therapy, PN= Picture Naming (personalized).
*L1: First acquired language (not necessarily dominant language).

Table 2
Summary Table tDCS-Related Effects.

	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
Aphasia type	nfvPPA	lvPPA	lvPPA	lvPPA	svPPA	NFPA	NFPA
First phase	Sham	Sham	tDCS	tDCS	tDCS	Sham	Sham
Change trained items (picture naming)							
Sham	↑	→	↑	→	→	↑	→
tDCS	↑	↑	↑	↑	→	↑	→
Change untrained items and tasks (treated language)							
Sham							
BNT	→	→	↓	↓	→	→	→
Sentence comprehension	→	→	→	→	→	→	→
Word repetition	→	→	→	→	→	→	→
Sentence construction	→	→	→	→	→	→	→
Semantic categories	→	→	→	→	→	→	→
tDCS							
BNT	↑	↑	↑	↑	→	↑	→
Sentence comprehension	→	↑	↑	→	→	→	→
Word repetition	→	↑	→	→	→	↑	→
Sentence construction	→	→	↑	→	→	→	→
Semantic categories	→	→	→	→	↑	→	→
Change untrained tasks (untreated language)							
Sham							
BNT	→	→	→	↑	↓	↑	→
Sentence comprehension	→	→	→	→	→	↑	→
Word repetition	→	→	→	→	→	→	→
Sentence construction	→	→	→	→	→	→	→
Semantic categories	→	→	→	→	→	→	→
tDCS							
BNT	↑	↑	→	→	→	↑	→
Sentence comprehension	↑	↑	→	→	→	↑	→
Word repetition	→	→	→	→	→	↑	→
Sentence construction	→	→	→	→	→	→	→
Semantic categories	→	→	→	→	→	→	→

↑ = significant improvement
↓ = significant decline
→ = no significant change
Note. Abbreviations: nfvPPA = non-fluent variant PPA, lvPPA = logopenic variant PPA, svPPA = semantic variant PPA, NFPS = non-fluent post-stroke. BNT = Boston Naming Test. Colour codes: green = significant improvement, red = significant decline, yellow = no significant change.

Table 3
Case 1 Patient Characteristics and Baseline Performance.

Aphasia type	nfvPPA-AoS	
Sex	Male	
Years of education (y)	10	
Age at diagnosis (y)	66	
Age at enrollment (y)	69	
MMSE (30)	29	
LANGUAGE	Dutch (L1)	French (L2)
Personalized picture naming (120)[1]	98	
Boston Naming Test (177)[2]	153	138
BAT Screening Part B	62/62 (100 %)	59/64 (92 %)
BAT Screening Part C	29/40 (73 %)	30/40 (75 %)
INHIBITORY CONTROL		
Stroop test (L1)		
Interference score (s)	75	
Percentile	31	
Attention Network Test		
Flanker effect (ms)	54.35	

Note. Abbreviations: MMSE = Mini-Mental State Examination, BAT = Bilingual Aphasia Test, RT = response time. Maximum test scores are listed within brackets.[1]Personalized picture naming consisted of 40 picture items and the [2]Boston Naming Test consists of 59 picture items. On both tests, items were scored on a 4-point scale and could receive 0,1, 2 or 3 points.

Table 4
Case 2 Patient Characteristics and Baseline Performance.

Aphasia type	lvPPA	
Sex	Male	
Years of education (y)	16	
Age at diagnosis	58	
Age at enrollment (y)	59	
MMSE (30)	19	
LANGUAGE	French (L1)	Dutch (L2)
Personalized picture naming (108)[1]		42
Boston Naming Test (177)[2]	65	97
BAT Part B	166/219 (76 %)	153/204 (75 %)
BAT Part C	28/43 (65 %)	25/43 (58 %)
INHIBITORY CONTROL		
Stroop test (L2)		
Interference score (s)	70	
T-score	40	
Attention Network Test		
Flanker effect (ms)	76.07	

Note. Abbreviations: MMSE = Mini-Mental State Examination, BAT = Bilingual Aphasia Test, RT = response time. Maximum test scores are listed within brackets.[1]Personalized picture naming consisted of 36 picture items and the [2]Boston Naming Test consists of 59 picture items. On both tests, items were scored on a 4-point scale and could receive 0,1, 2 or 3 points.

While AD and PPA are distinct, they share overlapping cognitive profiles, and the AD data serve as a useful benchmark in the absence of PPA-specific norms. This approach also applies to PPA Cases 2 to 5.

3.1.3. Effects of cerebellar tDCS on task performance

In summary, significant improvements were observed in the trained personalized picture naming task (L1) after both sham and anodal tDCS, with greater effects following anodal stimulation. Untrained BNT scores in L1 remained unchanged after both interventions, while L2 scores improved significantly only after anodal tDCS.

Performance on the **trained personalized picture naming task (L1)** improved significantly after sham, $\chi^2(1, N = 40) = 6.76$, $p = 0.009$, and even more so after anodal tDCS, $\chi^2(1, N = 40) = 31.84$, $p < 0.001$. **Untrained BNT scores** in L1 did not change significantly after sham,

Table 5
Case 3 Patient Characteristics and Baseline Performance.

Aphasia type	lvPPA	
Sex	Male	
Years of education (y)	10	
Age at diagnosis (y)	59	
Age at enrollment (y)	63	
MMSE (30)	26	
LANGUAGE	Dutch (L1)	French (L2)
Personalized picture naming (57)[1]		17
Boston Naming Test (177)[2]	n.a.	115
BAT Part B	73/206 (35 %)	172/240 (72 %)
INHIBITORY CONTROL		
Stroop test (L2)		
Interference score (s)	90	
z-score	0.58	
Attention Network Test		
Flanker effect (ms)	153.89	

Note. Abbreviations: MMSE = Mini-Mental State Examination, BAT = Bilingual Aphasia Test, RT = response time. Maximum test scores are listed within brackets.[1]Personalized picture naming consisted of 19 picture items and the [2]Boston Naming Test consists of 59 picture items. On both tests, items were scored on a 4-point scale and could receive 0,1, 2 or 3 points.

Table 6
Case 4 Patient Characteristics and Baseline Performance.

Aphasia type	lvPPA	
Sex	Male	
Years of education (y)	10	
Age at diagnosis (y)	69	
Age at enrollment (y)	73	
MMSE (30)	27	
LANGUAGE	French	Dutch
Personalized picture naming (63)[1]	24	
Boston Naming Test (177)[2]	70	63
BAT Part B	113/124 (91 %)	110/124 (89 %)
BAT Part C	27/41 (66 %)	32/41 (78 %)
INHIBITORY CONTROL		
Stroop Test (French)		
Interference score (s)	326	
Percentile	8	
Attention Network Test		
Flanker effect (ms)	157.53	

Note. Abbreviations: MMSE = Mini-Mental State Examination, BAT = Bilingual Aphasia Test, RT = response time. Maximum test scores are listed within brackets.[1]Personalized picture naming consisted of 21 picture items and the [2]Boston Naming Test consists of 59 picture items. On both tests, items were scored on a 4-point scale and could receive 0,1, 2 or 3 points.

$\chi^2(1, N = 59) = 0.89, p = 0.35$ or tDCS, $\chi^2(1, N = 59) = 1.19, p = 0.28$. In contrast **L2 BNT scores** increased significantly after tDCS, $\chi^2(1, N = 59) = 4.84, p = 0.03$, but not after sham, $\chi^2(1, N = 59) = 0.53, p = 0.47$. No significant changes were observed in **BAT scores** following either intervention.

A repeated measures ANOVA on **ANT** data revealed significant main effects of time, $F(1, 64) = 23.98$, $p < 0.001$, and condition, $F(1, 64) = 7.17$, $p = 0.009$, indicating a robust Flanker effect. A significant treatment × time interaction was also observed, $F(1, 64) = 4.252$, $p = 0.04$. Pairwise comparisons showed that this interaction was driven by superior baseline mean reaction times in the combined incongruent and congruent condition, during the tDCS phase (T3) compared to the sham phase (T1), with a difference of 133.15 ms, $F(1, 64) = 6.3$, $p = 0.02$). **Stroop *T*-scores** showed modest improvements following both sham (T1 *T*-score = 45, T2 *T*-score = 44) and tDCS (T3 *T*-score = 47, T4 *T*-score = 46).

A complete overview of pre- and post-treatment scores for Case 1 on all language and inhibitory tasks is provided in Supplementary table 2.

Table 7
Case 5 Patient Characteristics and Baseline Performance.

Aphasia type	svPPA	
Sex	Male	
Years of education (y)	6	
Age at diagnosis (y)	74	
Age at enrollment (y)	77	
MMSE (30)	12	
LANGUAGE	French (L1)	Dutch (L2)
Personalized picture naming (117)[1]	61	
Boston Naming Test (177)[2]	98	61
BAT Part B	148/231 (64 %)	81/201 (40 %)
BAT Part C	13/43 (30 %)	12/43 (28 %)
INHIBITORY CONTROL		
Stroop test (L1)		
Interference score (s)	434	
z-score	17.59	
Attention Network Test		
Flanker effect (ms)	146.76	

Note. Abbreviations: MMSE = Mini-Mental State Examination, BAT = Bilingual Aphasia Test, RT = response time. Maximum test scores are listed within brackets.[1]Personalized picture naming consisted of 39 picture items and the [2]Boston Naming Test consists of 59 picture items. On both tests, items were scored on a 4-point scale and could receive 0,1, 2 or 3 points.

Table 8
Case 6 Patient Characteristics and Baseline Performance.

Aphasia type Sex	Non-fluent post-stroke aphasia Male	
Years of education (y)	9	
Age at diagnosis (y)	74	
Age at enrollment (y)	77	
MMSE (30)	26	
LANGUAGE	French (L1)	Dutch (L2)
Personalized picture naming (81)[1]	25	
Boston Naming Test (177)[2]	132	105
BAT Part B	177/187 (95 %)	148/187 (79 %)
BAT Part C	28/43 (65 %)	24/43 (56 %)
INHIBITORY CONTROL		
Stroop test (L2)		
Interference score (s)	326	
Percentile	21	
Attention Network Test		
Flanker effect (ms)	58.61	

Note. Abbreviations: MMSE = Mini-Mental State Examination, BAT = Bilingual Aphasia Test, RT = response time. Maximum test scores are listed within brackets. [1]Personalized picture naming consisted of 27 picture items and the [2]Boston Naming Test consists of 59 picture items. On both tests, items were scored on a 4-point scale and could receive 0,1, 2 or 3 points.

3.2. CASE 2 – lvPPA

3.2.1. Participant characteristics and language background

Mr. G., a 59-year-old Belgian professor, began experiencing language difficulties upon returning to work after nearly two years of isolation during the COVID-19 pandemic. Referred to the hospital in early 2022, he reported issues with hearing and understanding spoken language, although the exact duration of symptoms was unclear. A neuropsychological evaluation conducted a year prior to inclusion revealed non-fluent, agrammatic speech with phonological and semantic paraphasias, word retrieval difficulties, and articulation issues. His language comprehension was mildly affected, and he showed deficits in episodic memory and visual encoding. Behaviorally, he reported decreased spontaneity, challenges with goal-oriented behavior and planning, and an increased appetite. Attention control was preserved, though he struggled with divided attention. Visuoconstruction abilities were intact. MRI showed mild hippocampal and global subcortical atrophy, and the core cerebrospinal fluid (CSF) biomarkers were consistent with Alzheimer's disease, supporting the diagnosis of lv PPA at age 58.

His L1 was French, spoken at home during childhood, and he learned Dutch (L2) at age 12, later becoming dominant in Dutch. At the time of enrolment, he primarily used L1 with family and **L2 during therapy**. SLT included personalized picture naming therapy (36 items), word-finding exercises, semantic categorization, and syllable reinforcement.

3.2.2. Baseline performance

3.2.2.1. Effects of cerebellar tDCS on task performance. In summary, Mr. G showed significant improvements in the trained personalized naming task after tDCS, with effects lasting until the 2-month follow-up. Untrained BNT scores improved significantly after tDCS in both L2 and L1, with lasting effects in L1. BAT syntactic comprehension and repetition scores in L2 also improved significantly after tDCS. ANT results showed reduced RTs in the incongruent condition following tDCS, with effects lasting until follow-up. However, Stroop interference scores declined slightly after tDCS.

Mr. G. showed significant improvement on the accuracy scores of the trained **personalized naming task in L2** after tDCS, $\chi^2(1, N = 36) = 26.70$, $p < 0.001$, but not after sham, $\chi^2(1, N = 36) = 14.73$, $p = 0.59$. The effect of tDCS persisted until follow-up, $\chi^2(1, N = 36) = 15.24$, $p < 0.001$. **Untrained BNT scores** improved significantly after tDCS in both the treated language (L2: $\chi^2(1, N = 59) = 19.70$, $p < 0.001$), and the untreated language (L1: $\chi^2(1, N = 59) = 4.33$, $p = 0.04$), but not after sham (L2: $\chi^2(1, N = 59) = 5.57$, $p = 0.18$; L1: $\chi^2(1, N = 59) = 0.18$, $p = 0.67$). The improvement in L1 lasted until follow-up, $\chi^2(1, N = 59) = 8.40$, $p = 0.001$. **BAT syntactic comprehension scores** improved significantly in L2 after tDCS, $\chi^2(1, N = 51) = 4.77$, $p = 0.03$, but not after sham, $\chi^2(1, N = 51) = 0.47$, $p = 0.3$. However, the significant improvement in syntactic comprehension did not persist until follow-up. **Repetition scores** showed a similar pattern, with significant improvement after tDCS, $\chi^2(1, N = 30) = 3.32$, $p = 0.03$, but not after sham, $\chi^2(1, N = 30) = 0.11$, $p = 0.37$.

A repeated measures ANOVA on ANT data revealed a significant main effect of congruency, $F(1, 92) = 63.27$, $p < 0.001$. After Bonferroni correction, a significant treatment × congruency × time interaction emerged, $F(1, 92) = 5.93$, p = 0.02, indicating a decreased mean RT after tDCS compared to baseline in the incongruent condition ($p = 0.04$), with effects lasting until follow-up ($p = 0.048$.) The treatment × time interaction was also significant, $F(2, 92) = 5.14$, $p = 0.02$, showing lower RTs in the incongruent condition at follow-up after tDCS compared to sham. The Flanker effect was significant at all time points. Mr. G.'s **Stroop** *T*-score improved after sham, increasing from 40 (16th percentile) to 47 (38th percentile), but declined after tDCS, decreasing from 46 (34th percentile) to 45 (31st percentile).

A complete overview of pre- and post-treatment scores for Case 2 on all language and inhibitory tasks is provided in Supplementary table 3.

3.3. CASE 3 – lvPPA

3.3.1. Patient characteristics and language background

Mr. V, a 63-year-old former communication director, first noticed symptoms seven years before participating in our study. These included difficulties with multisyllabic word pronunciation, word retrieval, and memory. At age 59, he was diagnosed with PPA, confirmed by FDG-PET imaging showing hypometabolism in the left superior and medial temporal cortex. He took leave from work and began bi-annual follow-ups. Initial neuropsychological testing showed oral production deficits, and two years later, follow-up assessment confirmed lvPPA, characterized by non-fluent speech, word retrieval difficulties, phonetic distortions and simplified syntax. Comprehension remained relatively preserved. SLT focused on lexical retrieval and phonological difficulties through multimodal naming, SFA, and Phonological Components Analysis.

Mr. V's L1 was Dutch, spoken at home and in school until age six. He then transitioned to a French-speaking school (L2), and over time, his dominance shifted to French, which he used with his wife and in his professional life. Before diagnosis, he reported equal proficiency in L1 and L2 (10/10 in all modalities), but post-diagnosis, he was mostly exposed to L2, and L1 deteriorated more rapidly. **SLT took place in his dominant language (L2)**, and included personalized picture naming therapy (19 items), phonological component analysis (Leonard et al., 2008), and syllable reinforcement.

3.4. Baseline performance

3.4.1. Effects of cerebellar tDCS on task performance

In summary, performance on the trained L2 naming task improved after both sham and tDCS, more so after tDCS, with greater gains following tDCS. Untrained naming in L2 declined after sham. Improvements in morphological opposites and sentence construction occurred post-tDCS. tDCS did not affect ANT inhibitory control but may have slowed down decline in verbal Stroop performance. No significant changes were found in untrained L1.

Mr. V. showed improvement on the **trained L2 picture naming task** following both sham, $\chi^2(1, N = 19) = 11.57$, $p < 0.001$, and tDCS, $\chi^2(1, N = 19) = 40.16$, $p < 0.001$, with a greater increase after tDCS (26/57 vs. 18/57 with sham). On the **BNT in L2**, performance did not significantly differ after tDCS, $\chi^2(1, N = 59) = 2.88$, $p = 0.09$, but declined significantly after sham, $\chi^2(1, N = 59) = 7.05$, p = 0.01, and continued to decline at follow-up, $\chi^2(1, N = 59) = 23.68$, $p < 0.01$. In L1, no significant decline was observed after sham, $\chi^2(1, N = 59) = 1.8$, $p = 0.18$, and we were not able to test BNT in L1 after tDCS. Post-hoc error analysis revealed reductions in omissions and phonological errors after both sham and tDCS. **On the BAT in L2**, syntactic comprehension scores improved at follow-up post-tDCS ($\chi^2(1, N = 38) = 7.118$, $p = 0.02$). Gains were also significant for morphological opposites, $\chi^2(1, N = 10) = 7$, $p = 0.01$ and sentence construction, $\chi^2(1, N = 32) = 4.5$, $p = 0.03$ after tDCS, but not after sham (respectively $\chi^2(1, N = 10) = 7$, $p = 0.08$ and $\chi^2(1, N = 32) = 4.5$, $p = 0.18$).

In the **Stroop** interference task, reaction time increased after tDCS (90 s to 105 s, *z*-scores: 0.58 to 1.36) and even more so after sham (101 s to 145 s, *z*-scores: 1.15 to 3.47). In the **ANT**, significant main effects were found for treatment, $F(1, 69) = 14.89$, $p < 0.001$ and condition, $F(1, 69) = 102.44$, $p < 0.001$, as well as a significant treatment × time interaction, $F(1, 68) = 12.93$, $p < 0.001$.

A complete overview of pre- and post-treatment scores for Case 3 on all language and inhibitory tasks is provided in Supplementary table 4.

3.5. CASE 4 – lvPPA

3.5.1. Patient characteristics and language background

Mr. C., a 73-year-old bilingual speaker of Dutch and French, was diagnosed with lvPPA 15 months before joining our study. He first noticed word-retrieval issues at age 68, which progressed over 4 years and led to a referral to a neurologist. Assessment showed word-finding difficulties, word substitutions, and calculation issues. Mr. C. scored 24/30 on the MMSE. He produced semantic and phonological paraphasias during naming. Comprehension remained preserved, but he showed deficits in working memory, clock-drawing, and executive functioning, along with impulsivity and difficulties with planning. Analysis of CSF biomarkers and brain perfusion SPECT imaging (see Supplementary Materials) suggested Alzheimer's disease as the underlying cause of lvPPA in this patient. Mr. C. reported using Dutch and French equally, with no clear dominant language. After diagnosis, SLT was delivered in French, focusing on semantic categories, fluency, and memory exercises. His family interactions were also primarily conducted in French.

3.5.2. Baseline performance

3.5.2.1. Effects of cerebellar tDCS on task performance. In summary, Mr. C. improved on the trained naming task in French only after tDCS, and showed generalization to untrained items in both languages following tDCS, with maintenance at follow-up in French. In contrast, performance declined after sham. No gains were observed on the BAT, and executive function outcomes were mixed, with slight deterioration in Stroop performance post-tDCS, and longer response times during sham in the ANT.

Mr. C. showed significant improvement on the **trained naming task** in French after tDCS, $\chi^2(1, N = 21) = 8.07$, $p = 0.01$, but not after sham $\chi^2(1, N = 21) = 1.8$, $p = 0.17$. On **the untrained BNT**, significant improvements were observed in both French, $\chi^2(1, N = 59) = 4.9$, $p = 0.03$, and Dutch, $\chi^2(1, N = 59) = 7.81$, $p = 0.01$), after tDCS. Scores in French remained stable at follow-up, $\chi^2(1, N = 59) = 7.36$, $p = 0.01$. After sham, Dutch scores remained unchanged, $\chi^2(1, N = 59) = 1$, $p = 0.32$, while French scores declined significantly, $\chi^2(1, N = 59) = 9.98$, $p < 0.01$. No improvements were observed on the **BAT**, and sentence construction scores declined at T3, $\chi^2(1, N = 32) = 4.5$, $p = 0.03$.

Stroop interference worsened following tDCS (*T*-score declined from 41 to 37), but improved after sham (*T*-score increased from 31 to 39). A repeated measures ANOVA on **ANT** data revealed main effects of intervention, $F(1, 66) = 5.52$, $p = 0.02$, and condition, $F(1, 66) = 43.43$, $p < 0.001$. Pairwise comparisons indicated longer mean RTs during sham, indicating performance decline. However, no significant effects in mean RTs were observed after tDCS.

A full overview of pre- and post-treatment performance on language and inhibitory tasks with raw scores for Case 4 is provided in Supplementary table 5.

3.6. CASE 5 – svPPA

3.6.1. Patient characteristics and language background

Mr. B, a 77-year-old retired chef, began noticing cognitive difficulties at age 74, including memory lapses, temporal disorientation, and word-finding problems. He scored 21/30 on the MMSE, with deficits in executive functioning, naming, and clock-drawing. MRI revealed supratentorial white matter lesions, while FDG PET-CT showed significant hypometabolism in several left hemisphere regions, including the superior parietal cortex, precuneus, angular cortex, posterior lateral temporal cortex, and frontal premotor cortex. No abnormalities were found in the sensory-motor cortex, basal ganglia, thalamus, or cerebellum. Initially diagnosed with Alzheimer's disease, he began treatment with Memantine and received biannual follow-ups. 2 years later, he was diagnosed with svPPA, scoring 48/100 on the ACE-revisited, with semantic impairment primarily accounting for the low score. Raised bilingually in French (L1) and Dutch (L2), Mr. B. was equally proficient in both languages, though L2 declined more after diagnosis. SLT in L1 included personalized picture naming (39 items) and exercises in verbal fluency and semantic categories.

3.7. Baseline performance

3.7.1. Effects of cerebellar tDCS on task performance

In summary, cerebellar tDCS yielded significant effects on semantic categories but not on naming tasks or the BNT in either language. ANT results showed robust Flanker effects, with performance declining over time, while Stroop performance worsened after tDCS but improved after sham.

No significant effects were found on the **trained picture naming task** following either tDCS, $\chi^2(1, N = 39) = 0.13$, $p = 0.71$, or sham stimulation, $\chi^2(1, N = 39) = 0.56$, $p = 0.46$. However, a significant decline occurred at follow-up after sham, $\chi^2(1, N = 39) = 8.07$, $p = 0.01$. Similarly, there were no significant effects on the **BNT** in L1 following either tDCS, $\chi^2(1, N = 59) = 2.13$, $p = 0.14$, or sham, $\chi^2(1, N = 59) = 0.10$, $p = 0.75$, although scores declined at follow-up after both tDCS, $\chi^2(1, N = 59) = 6.26$, $p = 0.01$, and sham, $\chi^2(1, N = 59) = 14.73$, $p < 0.001$. In L2, no significant changes were observed on the BNT after tDCS, $\chi^2(1, N = 59) = 1.33$, $p = 0.25$, or sham, $\chi^2(1, N = 59) = 0.33$, $p = 0.56$. However, a significant decline was noted at follow-up after sham, $\chi^2(1, N = 59) = 0.33$, $p = 0.57$, but not after tDCS, $\chi^2(1, N = 59) = 4.84$, $p = 0.03$. On the **BAT** in L1, there was significant improvement in semantic categories after tDCS, $\chi^2(1, N = 5) = 4.00$, $p = 0.046$, while syntactic comprehension scores declined significantly at follow-up after sham, $\chi^2(1, N = 5) = 4.00$, $p = 0.02$.

ANT results revealed main effects for intervention, $F(1,61) = 6.47$, $p = 0.01$, time, $F(1,60) = 3.90$, $p = 0.03$, and condition, $F(1, 61) = 48.07$, $p < 0.001$. A significant interaction effect was found for the intervention × condition, $F(1,61) = 4.82$, $p = 0.03$, indicating better performance on the incongruent condition in the tDCS phase compared to sham. Additionally, the time × condition interaction was significant, $F(1,60) = 3.26$, $p = 0.05$, showing a significant difference between T1 and T2 in the congruent condition. **Stroop** performance declined after tDCS (434 s to 476 s, with *z*-scores from 17.59 to 22.82), but improved in the sham phase, although initial performance was considerably worse (901 s to 825 s, with *z*-scores decreasing from 41.54 to 37.64; follow-up = 837 s, *z*-score = 38.26).

A full overview of pre- and post-treatment performance on language and inhibitory tasks for Case 5 is provided in Supplementary table 6.

4. Cases WITH POST-STROKE APHASIA

4.1. CASE 6 – Chronic non-fluent post-stroke aphasia

4.1.1. Patient characteristics and language background

Mr. J, a 72-year-old man, experienced an ischemic stroke three years before enrollment in our study. The stroke affected the left frontal gyrus, including the inferior and middle frontal regions and pars opercularis, due to a left middle cerebral artery infarct. He initially presented with expressive aphasia and facial nerve paresis. MRI revealed chronic infarctions in the left frontal lobe and right precentral gyrus. Raised bilingually in French (L1) and Dutch (L2), Mr. J. predominantly spoke French in early childhood, later becoming fluent in Dutch after entering a Dutch school from age 11. He reported a slightly better proficiency in French (10/10) than Dutch (9/10). Following the stroke, Mr. J received SLT in Dutch. His most recent assessment, nine months prior to the study, confirmed non-fluent aphasia, with marked difficulties in word fluency, naming, sentence comprehension, and picture description. SLT included personalized naming therapy (27 items), and word-finding exercises such as verbal fluency tasks.

4.1.2. Baseline performance

On the ANT, mean RTs were 1329.3 ms in the congruent condition and 1387.9 ms in the incongruent condition. While no normative data are available for the ANT in post-stroke aphasia, LaCroix et al. (2021) reported mean RTs of 1160.04 ms (congruent) and 1344.96 ms (incongruent) in 22 individuals with chronic post-stroke aphasia following a single left-hemisphere stroke (mean age: 55). The Stroop interference task was not reassessed after T1 due to time constraints during testing.

4.1.3. Effects of cerebellar tDCS on task performance

In summary, scores for trained naming in L2 and syntactic comprehension in L1 improved following both sham and anodal tDCS. In contrast, repetition, untrained naming in both L1 and L2, picture description in both languages, and ANT RTs (congruent condition) improved only after anodal tDCS.

Performance on the **trained** personalized **picture naming task** improved significantly after both sham, $\chi^2(1, N = 27) = 22.09$, $p < 0.001$, and anodal tDCS $\chi^2(1, N = 27) = 13$, $p < 0.001$, with slightly greater gains after anodal tDCS. Notably, the improvement remained significant at follow-up only after anodal tDCS $\chi^2(1, N = 27) = 4.57$, $p = 0.03$. **Untrained BNT** scores in L2 (the treated language) improved significantly only after anodal tDCS, $\chi^2(1, N = 59) = 16.2$, $p < 0.001$, and the gains were sustained at follow-up, $\chi^2(1, N = 59) = 7.12$, $p = 0.01$. Untrained BNT scores in L1 (the untrained language) also improved significantly after anodal tDCS, $\chi^2(1, N = 59) = 6$, $p = 0.01$. Word repetition scores in L1 remained unchanged after sham $\chi^2(1, N = 30) = 1$, $p = 0.32$, but improved significantly after anodal tDCS $\chi^2(1, N = 30) = 5$, $p = 0.03$. **BAT** syntactic comprehension scores in L1 improved significantly following both sham $\chi^2(1, N = 66) = 7.36$, $p = 0.01$, and anodal cerebellar tDCS $\chi^2(1, N = 66) = 7$, $p = 0.01$.

A repeated measures ANOVA on **ANT** data, revealed a significant interaction of congruency × stimulation × timing, $F(2, 54) = 12.4$, $p < 0.001$. After Bonferroni's correction, a significant effect was found exclusively in the anodal tDCS condition, specifically for the congruent condition, with significantly decreased RTs between T3 and T4, $F(2,54) = 23.58$, $p < 0.001$, and between T3 and T5, $F(2,54) = 23.58$, $p < 0.001$. A Flanker effect was evident at all timepoints except for T1 (baseline sham measurement).

A full overview of pre- and post-treatment performance on language and inhibitory tasks for Case 6 is provided in Supplementary table 7.

4.2. CASE 7 – Chronic non-fluent post-stroke aphasia

4.2.1. Patient characteristics and language background

4.5 years prior to inclusion in our study, at age 65, Ms. G. experienced a left frontotemporal and intraventricular hemorrhage due to rupture of the left sylvian artery. This resulted in right-side hemiplegia, hemianopsia, and non-fluent aphasia, characterized by perseveration and stereotypies. Despite lexical retrieval errors, her comprehension remained mostly intact. Eight months of rehabilitation led to gradual improvement in lexical comprehension, although difficulties with syntax and stereotypy persisted. She was attentive, regularly watched TV, and received speech therapy at home. Testing in French (L1) showed severe speech limitations with perseveration, and the following scores: 10/30 for repetition, 8/9 for word comprehension and 21/38 for sentence comprehension. Written production was not assessed due to hemiplegia, and aphasia precluded valid executive function testing. Ms. G. was raised by French-speaking parents in a Dutch-speaking city. Although she initially struggled with French literacy, she caught up after moving to a French-speaking environment from age 12. She rated her French at 8/10 and Dutch at 5/10. SLT was conducted in French and included personalized picture naming and sentence comprehension tasks (15 items each, see Table 9).

4.2.2. Baseline performance

The Stroop test was not conducted due to her severe aphasia, and the ANT was omitted because of her hemiplegia.

Table 9
Case 7 Patient Characteristics and Baseline Performance.

Aphasia type Sex	Non-fluent post-stroke aphasia Female	
Years of education (y)	6	
Age at diagnosis (y)	65	
Age at enrollment (y)	69	
LANGUAGE	French (L1)	Dutch (L2)
Naming trained items (45)[1]	12	
Sentence comprehension trained items (15)	11	
Boston Naming Test (177)[2]	9	6
BAT Part B	60/86 (70 %)	L2 55/86 (64 %)
BAT Part C	7/15 (47 %)	3/15 (20 %)

Note. Abbreviations: BAT = Bilingual Aphasia Test. Maximum test scores are listed within brackets.[1]Personalized picture naming consisted of 15 picture items and the [2]Boston Naming Test consists of 59 picture items. On both tests, items were scored on a 4-point scale and could receive 0,1, 2 or 3 points.

4.2.3. Effects of cerebellar tDCS on task performance

In summary, Ms. G. did not show significant improvements on any of the trained tasks following either sham or anodal cerebellar tDCS.

No significant changes were observed in performance on the **trained picture naming** task both sham stimulation, $\chi^2(1, N = 15) = 1$, $p = 0.32$, or after tDCS, $\chi^2(1, N = 15) = 1$, $p = 0.32$. Similarly, no significant improvement was found on the trained sentence comprehension task after sham, $\chi^2(1, N = 15) = 0.2$, $p = 0.65$, or tDCS, $\chi^2(1, N = 15) = 1$, $p = 0.32$. Ms. G. also did not show significant changes on the **BAT**.

5. Discussion

This case series of seven bilingual aphasia participants suggests that cerebellar tDCS may enhance the effects of SLT, including in various PPA subtypes (nfvPPA, svPPA, lvPPA) and chronic non-fluent post-stroke aphasia. Over 3 weeks, all participants received nine SLT sessions tailored to their language deficits, combined with sham and anodal cerebellar tDCS in a within-subject sham-controlled design. Treatment effects were assessed using trained and untrained naming tasks (BNT), BAT subtests, and inhibitory control tasks (ANT and Stroop). While both sham and tDCS produced positive outcomes for trained items, tDCS led to broader generalization effects, extending to untrained items, untreated tasks, and the untreated language. Long-term follow-up in Cases 2, 3, 4, 5, and 6 showed lasting effects in Cases 2, 4, and 6. To our knowledge, only our previously published case studies (Coemans et al., 2023; Coemans et al., 2024) have explored cerebellar tDCS in bilingual aphasia.

We first discuss SLT outcomes following sham stimulation, then analyze tDCS-related effects in trained tasks, untrained tasks and the untreated language. Our findings are contextualized in light of existing literature, with attention to potential mechanisms of cerebellar tDCS and its differing impact on the ANT and Stroop tasks. **Supplementary Materials** provide details on language impairment patterns in our bilingual sample.

5.1. Effects of speech and language therapy with sham tDCS

SLT improved trained picture naming in two PPA participants (Case 1, nfvPPA; Case 3, lvPPA) and one post-stroke aphasia participant (Case 6). Previous work has shown that SFA can facilitate lexical retrieval across PPA variants, although svPPA has been studied more extensively than nfvPPA (Jokel et al., 2009; Marcotte & Ansaldo, 2010) or lvPPA (Henry et al., 2019; Pagnoni et al., 2021). Post-hoc error analysis indicated that Case 1 improved primarily by reducing semantic and articulatory errors, while Case 3 made fewer phonological errors and omissions. These results support the benefit of combined semantic and phonological/phonetic training, independent of tDCS.

Case 1′s improvements demonstrate the effectiveness of semantic training in a participant with motor planning deficits, demonstrating that lexical retrieval can be retrained even with mild semantic impairments. While AoS was not explicitly targeted, correcting speech sound errors during SLT likely contributed to his improvement, consistent with evidence for treating nfvPPA-AoS (Henry et al., 2013; Jafari et al., 2018). Case 3′s gains are consistent with previous studies showing benefits of phonological approaches in lvPPA (Henry et al., 2019; Pagnoni et al., 2021). However, Case 5 (svPPA) did not improve in trained naming, despite prior findings showing that lexical retrieval treatment can be effective in svPPA (Beales et al., 2016; Croot et al., 2019; Dial et al., 2019; Harris et al., 2019; Lavoie et al., 2020; Meyer et al., 2018). This lack of improvement may reflect advanced semantic decline in svPPA, which reduces responsiveness to therapy unless explicitly addressed (Roncero et al., 2019).

Case 6 (post-stroke) showed significant improvements in trained naming after sham, including fewer omissions and faster RTs, aligning with previous reviews of SFA therapy in post-stroke aphasia (see Efstratiadou et al., 2018 for a review). Case 7, however, showed no improvement, likely due to a larger lesion, more severe aphasia, and limited baseline language production.

5.2. Effects of speech and language therapy with anodal tDCS

5.2.1. Trained tasks

Anodal tDCS elicited significantly larger effects than sham on semantic categorization in Case 5 (svPPA) and trained picture naming for Cases 1 (nfvPPA), 2 (nfvPPA), 4 (lvPPA) and 6 (post-stroke aphasia). Improvements generalized to untrained items (BNT) and were sustained at follow-up. Previous research has reported benefits of tDCS over *cortical* language areas in picture naming across PPA variants (e.g., Fenner et al., 2019; Nissim et al., 2022; Roncero et al., 2019; Tsapkini et al., 2014, 2018, Fenner et al., 2019; Ficek et al., 2018; Harris et al., 2019; Nissim et al., 2022; Roncero et al., 2019; Themistocleous et al., 2021). Our findings extend this literature by showing that *cerebellar* tDCS can similarly enhance language in PPA.

In Case 5 (svPPA), tDCS led to significant improvement on the BAT semantic categories subtest but not on trained naming, which is consistent with studies showing mixed results for naming in svPPA (Teichmann et al., 2016; Roncero et al., 2019). Teichmann et al. (2016), for example, reported benefits in semantic association tasks after tDCS targeting the anterior temporal lobe. Case 5 was the only PPA participant who did not show improvement in trained picture naming after tDCS, which is consistent with previous research suggesting that naming gains in svPPA following tDCS are often limited or inconsistent. While some studies have reported modest benefits from stimulating left temporoparietal regions (Roncero et al., 2019), these effects are generally weak. Hung et al. (2017) found more positive outcomes. Other work, such as Tsapkini et al. (2018), showed improvements following stimulation of the left IFG, but these did not generalize to untrained items. Moreover, several studies have reported group-level effects without distinguishing between PPA variants (Ficek et al., 2018; Harris et al., 2019; Roncero et al., 2019), making it difficult to draw conclusions about the effectiveness of tDCS in svPPA.

Improvements in semantic categories for Case 5 after cerebellar tDCS, but not in trained picture naming tasks, can be attributed to several factors. Semantic category tasks are generally less demanding in terms of lexical retrieval compared to picture naming tasks, which require precise word retrieval and are particularly challenging for individuals with svPPA due to their semantic deficits. Additionally, Case 5′s cortical atrophy may have affected regions involved in lexical retrieval, limiting the efficacy of tDCS. The extent of atrophy might have hindered the excitatory and metabolic changes typically induced by tDCS (de Aguiar et al., 2020; Nissim et al., 2020). Case 5 also exhibited significant inhibitory control deficits, as evidenced by performance on the Stroop and ANT tasks. While executive function deficits in PPA, especially in svPPA, have been debated (Macoir et al., 2017), our findings supports their presence, consistent with our previous meta-analysis (Coemans et al., 2022). These findings are consistent with previous research linking executive dysfunction to limited language recovery in post-stroke aphasia (Fridriksson et al., 2006; Mooijman et al., 2022).

In post-stroke aphasia, tDCS has shown potential for producing generalized, long-lasting improvements across language domains (Ding et al., 2022; Harvey & Hamilton, 2022), as illustrated by Case 6. Specifically, cerebellar tDCS has been associated with improved picture naming performance compared to sham stimulation (Marangolo et al., 2018; Sebastian et al., 2020, 2017). However, Case 7 did not show improvements, highlighting the individual variability in response to tDCS in chronic post-stroke aphasia (Baker et al., 2010; Ding et al., 2022; Elsner et al., 2019; Marangolo et al., 2013; Shah-Basak et al., 2015). Variables such as lesion size and location (Basilakos et al., 2014; Kang et al., 2024; Naeser et al., 1989; Norise & Hamilton, 2017), as well as compromised white matter integrity in Case 7 (Zhao et al., 2021), may have attenuated effects of stimulation. Comorbid conditions, including Case 7′s morbid obesity, may also influence treatment outcomes (Vergallito et al., 2022). These contrasting results emphasize the need for personalized treatment approaches in chronic non-fluent aphasia (de Aguiar et al., 2015).

5.2.2. Untrained tasks

Case 3 exhibited improved syntactic comprehension at follow-up, which aligns with previous studies suggesting that the effects of cerebellar tDCS may emerge at later time points (Maldonado et al., 2023; Spielmann et al., 2017). In post-stroke aphasia (Case 6), tDCS enhanced performance in untrained tasks, such as repetition and syntactic comprehension, supporting prior evidence for generalization of treatment effects (Biou et al., 2019; Wang et al., 2023).

5.2.3. Treatment language and cross-linguistic transfer (CLT) effects

The effectiveness of SLT continues to be a topic of debate, particularly regarding the influence of relative proficiency levels across languages (Goral & Lerman, 2020; Goral et al., 2023). Some researchers propose that activating a less proficient or later-acquired language may facilitate transfer to a stronger L1, in line with the Revised Hierarchical Model of bilingual lexical representation (Kroll & Stewart, 1994; Kroll et al., 2010). Conversely, suppressing a more dominant language to activate a weaker one may hinder cross-language generalization from L2 to L1 (Lerman et al., 2022).

Our findings indicate that SLT was effective in both PPA and post-stroke aphasia participants, regardless of whether the treated language was L1, L2, dominant, or non-dominant. CLT effects were observed following cerebellar tDCS in Cases 1, 2, 4, 6, extending to the untreated language on the BNT, an effect not seen after sham stimulation.

Case 6 showed CLT effects after both sham and tDCS, although only tDCS led to additional gains in syntactic comprehension and repetition.

Although CLT remains underexplored, it has been documented in both post-stroke aphasia (Ansaldo & Ghazi Saidi, 2014; Faroqi-Shah et al., 2010; Goral et al., 2023) and PPA (Henry et al., 2019; Meyer et al., 2015). Factors such as pre- and post-morbid language proficiency, structural similarities between languages and cognitive control (Ansaldo & Saidi, 2014; Goral et al., 2007) are thought to influence CLT. Treatment effects were generally stronger within the treated language, consistent with post-stroke bilingual aphasia literature (Goral et al., 2023). To our knowledge, this is the first study to show that cerebellar tDCS may enhance CLT.

5.2.4. How does cerebellar tDCS exert its effects?

Our study posits that cerebellar tDCS-related improvements in language performance may be explained by three potential mechanisms.

First, the efficacy of tDCS in tasks such as picture naming, sentence comprehension and production, and repetition, may derive from its ability to engage the residual left cortico-subcortical network via reciprocal white matter pathways connecting the cerebellum to key language-related cortical areas, including the prefrontal cortex, inferior frontal gyrus, superior temporal gyrus, and parietal cortex (Schmahmann, 2001). This facilitates bidirectional communication between the cerebellum and cortical regions involved in language processing (Bruffaerts et al., 2020, Indefrey & Levelt, 2000; Jobson et al., 2022; Price, 2010). Cerebellar tDCS is thought to influence Purkinje cell polarization (Chan & Nicholson, 1986; Parazzini et al., 2014), thereby altering activity in the deep cerebellar output nuclei (Rahman et al., 2014), and impacting cortical excitability and plasticity (Nitsche & Paulus, 2000; Priori et al., 1998), potentially facilitating rehabilitation (Lammers et al., 2024; Marangolo et al., 2020; Stoodley & Schmahmann, 2010). Anodal stimulation is thought to increase Purkinje cell firing, enhancing inhibitory input to cortical areas, while cathodal stimulation reduces this inhibition, thereby increasing cortical excitability (Grimaldi et al., 2016). However, the cerebellum's unique neuronal structure may result in polarity effects different from those seen with cortical tDCS.

Although previous studies reported mixed results regarding cathodal versus anodal cerebellar tDCS (Rahman et al., 2014; van Dun et al., 2016), our findings align with evidence supporting beneficial effects of anodal cerebellar tDCS (Sebastian et al., 2017; Turkeltaub et al., 2016). Positive results have also been observed with cathodal (Marangolo et al., 2018) and polarity-independent stimulation (Sebastian et al., 2020). Neuroimaging studies evidence structural and functional reorganization supporting language rehabilitation in both PPA (Henry et al., 2018; Mandelli et al., 2018; Wilson et al., 2016) and post-stroke aphasia (Guo et al., 2019; Shah-Basak et al., 2022). tDCS may modulate functional connectivity between the stimulation site and distant language-related areas (Ficek et al., 2018; Macher et al., 2013; Tao et al., 2021; Turkeltaub et al., 2016; Zhao et al., 2021). SLT may activate the remaining language network and promote compensatory functional reorganization across broader networks, including executive function areas, which could further be supported by cerebellar tDCS (Beeson et al., 2011; Dressel et al., 2010; Ficek et al., 2018; Jokel et al., 2016; Marcotte & Ansaldo, 2010; Sonty et al., 2003).

Second, tDCS-related improvements in language tasks may be attributed to enhanced cerebellar involvement in phonological processing. Although the cerebellum's exact role in this domain is still being explored, both neuroimaging (Marvel & Desmond, 2016; Peterburs et al., 2019; Sheu & Desmond, 2022; Turker et al., 2023), and lesion studies (Justus et al., 2005; Mariën et al., 2014; Silveri, 2021), support its contribution. Phonological processing is essential for mapping sound to higher-order linguistics representations and for storing verbal information in working memory (Baddeley, 1992). In picture naming, it supports retrieval and assembly of phonological word forms (Foxe et al., 2021; Gorno-Tempini et al., 2011), while in sentence comprehension, it aids in phonological decoding and segmentation (Baddeley et al., 1998; Lauro et al., 2010). Functional imaging studies have shown that the right cerebellum contributes to response selection and the retrieval of linguistic alternatives (Desmond et al., 1998), supporting its involvement in phonological processing and verbal working memory during sentence construction (Marvel & Desmond, 2016; Peterburs et al., 2019). Phonological processing also underlies grammatical structure, proper pronunciation and sentence coherence (Myers & Robertson, 2015). Additionally, phonological working memory helps with tasks that require memorizing and manipulating verbal sequences, such as the instructions in the BAT sentence construction task. The cerebellum's involvement in these processes suggests that its modulation through tDCS could enhance language performance in individuals with phonological deficits, as observed in Case 3 (lvPPA), 5 and 6 (nfvPPA) (Beales et al., 2019; Dial et al., 2020; Foxe et al., 2021;; Leyton et al., 2014). Although cerebellar tDCS has been shown to impact verbal working memory in healthy participants (Boehringer et al., 2013; Ferrucci et al., 2008; Macher et al., 2013), our study is the first to show these effects in individual with aphasia.

Third, the cerebellum's involvement in error detection (Runnqvist et al., 2016), prediction (Mariën & Manto, 2018; Moberget et al., 2014), motor control (Peach & Tonkovich, 2004), and timing may also explain its role in language tasks such as naming (Argyropoulos, 2016), sentence construction (Silveri, 2021) and comprehension (Hull, 2020; Lesage et al., 2017). It supports the integration of visual processing, semantic retrieval, and motor planning in picture naming, and helps in detecting and correcting articulatory errors. Impairments in these processes can result in speech errors and disfluency. During sentence comprehension, the cerebellum may help anticipate upcoming words or syntactic structures (D'Mello et al., 2017; Geva et al., 2021). Similarly, in sentence construction, it supports motor planning and execution, to ensure fluent and the accurate speech output (Geva et al., 2021; Hartsuiker & Kolk, 2001; Runnqvist et al., 2016). Although some studies emphasize the cerebellum's role in prediction during language comprehension (Bongaerts et al., 2022), our findings suggest its contributions likely extend beyond prediction alone, engaging additional mechanisms as outlined earlier.

5.3. Effects of cerebellar tDCS on inhibitory control

5.3.1. tDCS-related improvements on the attention network test

Cerebellar tDCS was associated with improved inhibitory control, as evidenced by significant changes on the ANT, a nonverbal measure. Case 2 (nfvPPA) demonstrated enhanced performance in the incongruent condition following tDCS, while Case 6 (NFPS), showed reduced mean RTs in the congruent condition, reflecting improvements in monitoring abilities (Struys et al., 2019). These findings indicate that cerebellar tDCS may positively affect broader cognitive functions, supporting the domain-generality of cognitive control mechanisms (Bialystok et al., 2004; Blumenfeld et al., 2016; Costa et al., 2008; Struys et al., 2019). The cerebellum's involvement in general behavioral control (Abutalebi & Green, 2016; Green & Abutalebi, 2013) and its structural and functional connections with cortical areas involved in cognitive control (Aron et al., 2007) further substantiate these effects. This contributes to a growing body of literature indicating that cerebellar tDCS can modulate monitoring (Wynn et al., 2019) and inhibitory control (Mannarelli et al., 2019; Mannarelli et al., 2020). Interestingly, both Cases 2 and 6 also exhibited more extensive language improvements, either within the same language (Case 2) or across languages (Case 6), possibly due to enhanced inhibitory and monitoring capacities. These effects were not observed after sham stimulation. No other participants demonstrated changes on the ANT after either sham or anodal tDCS. Future research should investigate the factors influencing cerebellar tDCS efficacy, including brain anatomy, atrophy and task demands. Of note, contrasting results were observed in Stroop performance, which are addressed in the next section.

5.3.2. tDCS-related improvements on the stroop test

Cerebellar tDCS effects on inhibititory control, as measured by the Stroop test, were variable and showed no consistent pattern. Case 1 (nfvPPA) improved after both sham and tDCS, while Case 3 (lvPPA) declined under both conditions. Case 2 (nfvPPA) and Case 4 (lvPPA) improved after sham but declined after tDCS, while Case 5 (svPPA) showed the reversed pattern.

These differences may stem from the distinct cognitive demands of the tasks. While both tasks engage inhibitory control regions (Backes et al., 2011; Okayasu et al., 2023), the Stroop test also recruits language processing areas (Okayasu et al., 2023), which could influence tDCS effects. However, the lack of consistent language-related improvements across measures suggests other factors are involved. Discrepancies between Stroop and ANT results indicate that cerebellar tDCS may affect not only inhibitory control, but also other processes such as motor coordination, attention, or sensorimotor integration. These broader influences reflect the cerebellum's integrative rol across multiple functional networks. This interpretation aligns with prior research indicating that stimulation effects are task- and difficulty- dependent (Ferrucci et al., 2008; Macher et al., 2013; Miniussi et al., 2013). Additionally, some studies found no effects of cathodal cerebellar tDCS on tasks like the N-back (van Wessel et al., 2016), and Stroop (Maldonado et al., 2019).

While both the ANT and Stroop task involve conflict resolution, they may rely on different mechanisms. According to Li et al. (2014), the Stroop task involves a semantic conflict between word meaning and ink color, whereas the Simon-like conflict in the ANT is spatial in nature. The Stroop effect is thought to emerge earlier in processing, i.e., at the stimulus identification stage, while the ANT primarily engages response selection. This distinction may account for the different effects observed across tasks (Li et al., 2014; Scerrati et al., 2017). Thus, while both tasks assess aspects of response inhibition, their differential sensitivity to cerebellar tDCS may be attributable to the specific nature of interference involved. These findings underscore the importance of considering task-specific characteristics when evaluating the cognitive effects of cerebellar stimulation.

Our study also differs from previous work by including individuals with PPA, a population characterized by high intra-individual variability in task performance due to underlying neurodegeneration. Such variability has been observed across tasks and time points (Dixon et al., 2007; Halliday et al., 2018), including the Stroop task (Duchek et al., 2009). This variability may complicate the interpretation of tDCS effects on complex tasks like the Stroop test, and suggests that the Stroop test might not be the most sensitive measure to detect the cognitive impact of cerebellar tDCS in this population.

5.4. Limitations and future directions

Our study on cerebellar tDCS in bilingual aphasia has several limitations. The small sample size and variability in symptoms and etiologies necessitate cautious interpretation, as these factors may have influenced results. Additionally, while we relied on detailed clinical documentation, we lacked imaging data or biomarker information for some participants, limiting insights into neural mechanisms and disease progression. Most cognitive assessments, except for the personalized naming task, were conducted only once pre- and post-intervention. Multiple baseline testing would have improved reliability, particularly in dementia populations where performance can fluctuate (as seen in the Stroop test). The MMSE, while widely used, could be complemented by other tools (e.g. fatigue or motivation questionnaires), to provide a more comprehensive assessment of cognitive status. The BAT, while useful for bilingual aphasia assessment, featured unequal item sets across languages, limiting statistical comparisons. The absence of pre-and-post intervention neuroimaging also restricted our ability to evaluate neural changes associated with tDCS. The male-dominated sample (7 out of 8 participants) may affect generalizability of our findings (Licata et al., 2023), and further research should aim for greater gender balance. All participants received SLT alongside tDCS, but session intensity and frequency varied. Although the intended protocol involved nine sessions over three weeks, scheduling conflicts extended the duration to four weeks for some participants. The individualized nature of SLT meant that therapy content was adapted to individual needs, with occasional input from the research team, potentially introducing variability across cases.

Future research should explore the differential effects of cathodal vs. anodal cerebellar tDCS, examine task-specific effects across cognitive domains, and investigate its impact across aphasia subtypes and severity levels. Incorporating consistent follow-up assessment and neuroimaging would support the development of personalized interventions. Although this study focused on bilingual participants to explore the interplay between tDCS, bilingualism, and language recovery, including monolingual patients as a contrast group could clarify whether the observed effects are specific to bilingual contexts, or more broadly applicable.

6. Conclusions

This case series investigated the effects of cerebellar tDCS in seven individuals with bilingual PPA or post-stroke aphasia. Our findings suggest that cerebellar tDCS may positively impact language abilities across aphasia types, including improvements in picture naming, sentence comprehension, production, and repetition. Additionally, two participants demonstrated improvements on the ANT, suggesting potential benefits for cognitive control and attentional processes. In contrast, Stroop task results were more variable, highlighting the complexity of executive functions outcomes. Notably, similar effects were observed in both PPA and post-stroke aphasia, which is promising for neurodegenerative disorders where language typically declines over time. Future research, including neuroimaging to assess structural and functional brain changes, is necessary to confirm these preliminary findings and to evaluate long-term outcomes. Investigating the mechanisms through which cerebellar tDCS influences both language and executive function networks will be critical for understanding its therapeutic potential.

Overall, this study contributes to the growing body of evidence supporting the use of non-invasive brain stimulation techniques, such as tDCS, in aphasia rehabilitation and highlights the cerebellum as a potential target area for therapeutic intervention.

CRediT authorship contribution statement

Silke Coemans: Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Esli Struys:** Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Kyrana Tsapkini:** Writing – review & editing. **Vânia de Aguiar:** Writing – review & editing, Resources. **Sebastiaan Engelborghs:** Writing – review & editing, Resources. **Jean-Christophe Bier:** Writing – review & editing, Resources. **Philippe Paquier:** Writing – review & editing. **Stefanie Keulen:** Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

We would like to express our sincere gratitude to the participants, their caregivers, and the speech and language therapists for their invaluable time, effort, and trust in this study. Their dedication and cooperation made this research possible, and we deeply appreciate their contributions.

Funding

Silke Coemans's research is supported by the Research Foundation-Flanders (FWO, Belgium) under grant numbers FWOAL938-Junior Research Project and FWO1290325N-Postdoctoral Mandate. Vânia de Aguiar's work is funded by the project "Verb processing and verb learning in children with pediatric posterior fossa tumors" (Project number VI.Vidi.201.003) within the NWOTalentprogramma Vidi SGW 2020, financed by the Dutch Research Council (NWO, the Netherlands). Kyrana Tsapkini's research is supported by grants from the Science of Learning Institute at Johns Hopkins University, as well as NIH/NIA (R01 AG068881, R01AG075404, R01AG075111), United States.

Relationships.

There are no additional relationships to disclose.

Patents and Intellectual Property.

There are no patents to disclose.

Other Activities.

There are no additional activities to disclose.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bandl.2025.105617.

Data availability

Data will be made available on request.

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